

Algorithm Benchmarking and Outcomes

**The Society Connectedness & Health Lab
Yale School of Public Health**

Church Closures Dashboard Project



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Review of the 2023 Format Raw Data

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Overview

The project objectives were as follows:

1. **REVIEW:** Evaluate the raw data and previously outlined cleaning, validation, and harmonization methods employed by Dr. Insang Song for accuracy and identify opportunities for improvement.
2. **PREPARATION:** Implement identified improvements to prepare the dataset for analysis and distribution to qualifying collaborators.
3. **METRICS:** Calculate the metrics necessary for dynamic visualization.

Additionally, the project Principal Investigator (PI), Professor Yusuf Ransome, requested the pipeline be reproducible for future collaborators and readily adaptable to potential changes in project scope.

Following receipt of the raw data in Summer 2025, a review was conducted and findings were compared against the original methodology provided by Dr. Song. When an updated dataset was subsequently delivered in May 2026, the pipeline was expanded to accommodate two designated variations: the **2023 Format** and the **2026 Format**.

This document evaluates the three sets of quality control assessments collected in parallel with the 2026 Format data cleaning and validation process outlined in **Step 2**. These assessments correspond to the three core phases of **Step 2**:

- a. Validating and consolidating unique addresses represented in the data
- b. Address-based geocoding

- c. Point-in-polygon spatial joining to annotate the various census boundaries

Important Considerations

Selected code excerpts are referenced below. The complete code used to generate this report can be found in [“Code/Process Data Update.R”](#).

Ignored Files

Individual-level files are withheld from GitHub and Git tracking in accordance with the Data Use Agreement (DUA) with the data provider. Many of these are stored in `**/KEEP LOCAL/` directories or are explicitly listed under the “Project-Specific Exclusions” section of the `.gitignore` file.

Most files are available on the SOCAH Lab OneDrive for the project, while others are user-specific and do not need to be shared. If there are local-only files for this project, they will be added to the “Church-Closures-Dashboard GitHub LOCAL ONLY Files” directory. This directory will contain `README.md` files specifying the intended use and filepath for each file.

If you do not have access to the SOCAH Lab OneDrive, carefully follow the codebase instructions to establish the repository and replicate results on your own device. Raw data access instructions may vary depending on your institution.

Environment Reproducibility

Results were produced using R (v4.5.2), RStudio (v2026.01.1+403), and Quarto (v1.8.27). The `renv` package (v1.1.4) is included to reproduce the coding environment, capturing all relevant packages and their versions. If issues arise when running the scripts, verify that the environment is initialized and that the correct versions of R, RStudio, and Quarto are being used.

Compatibility

Updating packages, R, or Quarto may introduce compatibility issues with legacy code that will need to be resolved on a case-by-case basis.

First-time users should initialize the environment to install all required packages and ensure a reproducible coding environment. To do so:

1. In the command-line application (i.e., Terminal for Macs or Git Bash for Windows) navigate to the codebase file location.

Command-Line Interface

```
cd "FILE-PATH/Church-Closures-Dashboard/"
```

2. Launch the project by opening the *.Rproj file in RStudio.
3. In the R console, activate the environment by running the following lines of code:

Command-Line Interface

```
renv::restore() # Download packages and their version saved in the  
↪ lockfile.
```

i Note

If you are asked to update packages, say no. `renv` is intended to recreate the same environment under which the project was created, making it reproducible. You are ready to proceed when running `renv::restore()` gives the output:

RStudio Console

```
- The library is already synchronized with the lockfile.
```

If you experience any trouble with this step, you might want to confirm that you are using R (v4.5.2) in the RStudio (v2026.01.1+403) and Quarto (v1.8.27). You can also read more about `renv` in their [vignette](#)¹.

Special Considerations for PO Boxes and Missing `address_line_1` Entries

The raw data review of the 2023 and 2026 Formats revealed several notable special cases requiring specific handling. These are described in the respective sections of Step 2: Clean and Validate Addresses and Geolocation, and Annotate Records with Census Boundaries. Notably, approximately 14% of businesses had at some point filed under a PO Box, and approximately 3% had at some point submitted a record with no `address_line_1` value (`address_line_1 == NA`; no blank strings were observed). For context, each entry in the dataset represents a unique combination of address and year over 1,080,764 unique ABI, with no year repeated within a given business.

Later in the analysis, religious organizations are classified as closed if they did not file for four consecutive years during the observed period; organizations that subsequently resumed filing are classified as reopened. Closures are further differentiated into two categories: those that stopped operations entirely and closures due to relocation. Relocated organizations remained operational but moved outside the community, thereby resulting in a loss of social benefits to that community.

Addresses filed under a PO Box or missing an `address_line_1` value present challenges for generating these metrics. Dr. Insong's method addressed this by removing all PO Box entries while retaining the remaining addresses for each business. However, since reported dates were unique across entries, this was expected to skew results. The Summer 2025 prototype methods sought to associate each PO Box with a nearby physical address (both spatially and temporally) to serve as a proxy for the organization's location on those dates. This relied on the assumption that PO Boxes are opened near the business's operating location, which was later deemed too strong an assumption to adopt.

For these reasons, under the following three conditions, all entries associated with a business are removed:

- Geocoordinates are absent and could not be recovered through address-based validation, preventing spatial joining (___).
- `address_line_1` is missing, preventing verification that the geocoordinates correspond to the institution itself rather than a general location within that ZIP code (3% ABI).
- PO Box used: the physical location of the business, and consequently its presence or absence within the community of impact, is ambiguous (14% ABI).

Table 1 shows the distribution of ABIs impacted by any one of these conditions. Not shown are results indicating that longitude and latitude were always missing together, affecting 0.4% of ABIs. These results demonstrate very little overlap between conditions, suggesting that approximately 17% of ABIs, nearly one-fifth of ABI, could potentially qualify for removal.

Table 1: Percentages of ABI with Address and Geolocation Considerations

NA Address = FALSE			NA Address = TRUE		
NA Lon	PO Box		NA Lon	PO Box	
	FALSE	TRUE		FALSE	TRUE
FALSE	83.51	13.48	FALSE	2.38	0.24
TRUE	0.36	0.02	TRUE	0.02	0.00

Simply removing ABIs with these characteristics introduces a potential source of bias, especially among those filing under a PO Box. Table 2 summarizes the share of ABIs affected by PO Box filings and missing `address_line_1` across U.S. cities, with some reaching 100% of ABIs impacted. Figure 1 maps these values spatially, confirming that some communities are disproportionately affected by this data loss. Note that 10.6% and 12.6% of cities could not be matched to TIGER/Line Shapefile features imported via the `tigris` package.

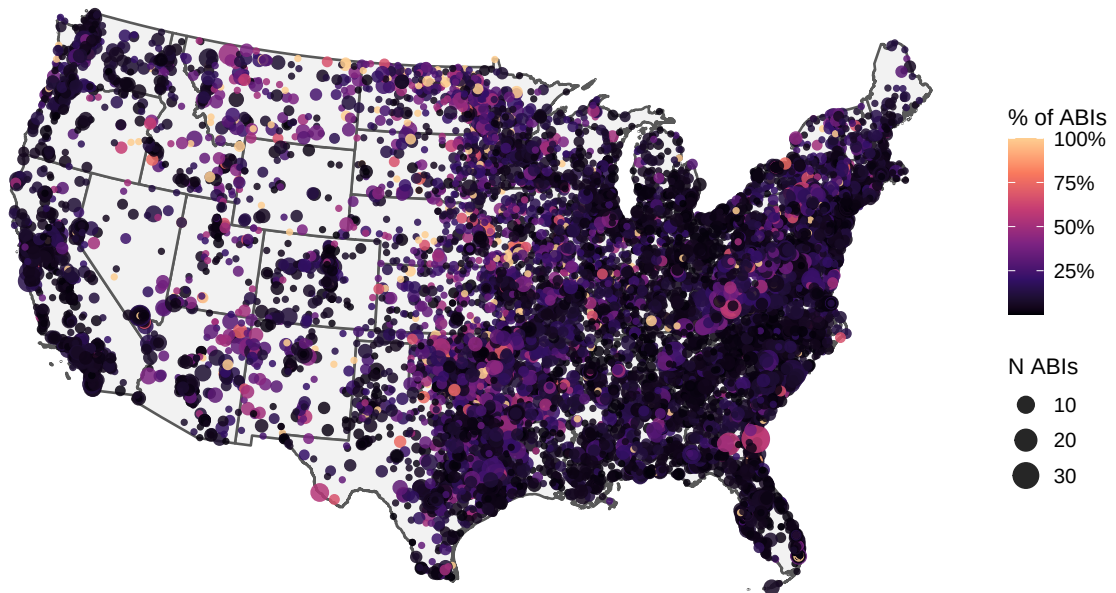
Table 2: Summary Statistics of ABIs with Select Characteristics by City

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
NA address line 1 (%)	0.09	2.56	6.25	14.93	17.65	100.00
PO Box (%)	0.26	15.38	33.33	42.16	63.64	100.00

The agreed-upon mitigation strategy is to remove an ABI only when the selected date range includes an address with a missing `address_line_1`, a PO Box filing, or an address without geocoordinates and, consequently, no census boundary mapping. The number of ABIs removed under each condition will be reported on the dashboard to inform users of the extent of data loss attributable to each condition.

ABIs with Any Missing Address Line 1 by City

Color = % of ABIs; size = N ABIs



ABIs using PO Boxes by City

Color = % of ABIs; size = N ABIs



Figure 1

Address Validation and Consolidation Process

QC1: API Usage, Matching Overview, and Unique Result Consistency

Address-Based Geocoding

QC1: US Census Bureau Geocoder API Validation

QC2: Reported vs. Validated Geocoordinate Comparison

QC3: Geocoordinate Variation by Address

Census Boundary Annotation Using Point-in-Polygon Spatial Joins

QC1: GEOID Vintage Alignment

QC2: Core Statistics Vintage Alignment

Appendix

State-Level Shapefile for Basemap

For the implementation, refer to **SUBSECTION A2: Distribution of Characteristics Requiring Removal** in “[Code/Clean Raw Data_Step 1_2026 Format.R](#)”. State-level census boundary files were incorporated using the `tigris` package ². Results were written to “./Data/Results/From Clean Raw Data/Step 2_2026 Format/us_states_sf.shp”.

Table 3: Codebook for us_states_sf 2023 Cartographic Boundaries Basemap

Field	Description
STATEFP	Two-digit state FIPS code (character).
STATENS	GNIS feature identifier for the state (character).
GEOIDFQ	Fully qualified geographic identifier (character).
GEOID	State geographic identifier (typically the state FIPS code as character).
STUSPS	USPS state abbreviation (e.g., “CA”) (character).
NAME	State name (character).
LSAD	Legal/statistical area description code (character).
ALAND	Land area in square meters (numeric/integer).
AWATER	Water area in square meters (numeric/integer).
geometry	State boundary geometry as MULTIPOLYGON.

Missing Address Lines and PO Boxes By City

For the implementation, refer to **SUBSECTION A2: Distribution of Characteristics Requiring Removal** in “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. The **Step 1** standardized data were summarized by city (as recorded on the form or, when needed, validated via ZIP code using the SimpleMaps Basic [United States Cities Database](#) (version 1.90) lookup table created with the custom function `build_zip_city_lookup()` ³). These results were then joined to city geocoordinates from the `tigris` package ². Results showing missing `address_line_1` and the presence of PO Boxes were written separately to “./Data/Results/From Clean Raw Data/Step 2_2026 Format/ABI with

NA Addresses by City_08.07.2026.csv" and
"/ABI with PO Boxes by City_08.07.2026.csv".

Table 4: Codebook for Evaluation Summary Output Fields (Missing address_line_1 and PO Box Datasets)

Field	Description
state/city/zip_code	Address elements associated with each entry.
n_abi	Number of unique ABIs represented in that city or zip code.
[n pct]_abi_any_na_address_line_1	Count and percent of ABIs with at least one missing address_line_1.
[n pct]_abi_poBox	Count and percent of ABIs with at least one PO Box entry.
lon/lat	TIGER/Line Shapefile geocoordinates associated with the city.

GEOID Prebuild Shapefiles

For the implementation, refer to **SUBSECTION A4: Build Precompiled TIGER/Line GeoPackages** in “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. As described in the methods for LOOP PART D: Point-in-Polygon Spatial Assignment of Census Information, pulling in shapefiles using the `tigris` package was computationally prohibitive. Instead, relevant state-level shapefiles at the block and block-group level were manually downloaded from the Census Bureau TIGER/Line Shapefiles database (see [Directory Specific Notes](#) in the project GitHub repository for direct links to sources)⁴. These were then compiled into standardized shapefiles with layers for each decennial period (2000, 2010, and 2020) and written to the “./Data/Results/Census Bureau TIGER Line Shapefiles” directory.

Table 5: Codebook for State Block and Block-Group Shapefile Layers (2000/2010/2020)

Field	Description
<code>decennial_year</code>	The decennial census year associated with the boundary definition (2000, 2010, or 2020).
<code>geoid</code>	Full concatenated census geographic identifier, combining the state, county, tract, and block-group FIPS codes into a unique identifier for each boundary unit.
<code>statefp</code>	Two-digit state Federal Information Processing Series (FIPS) code identifying the state.
<code>countyfp</code>	Three-digit county FIPS code identifying the county within the state.
<code>tractce</code>	Six-digit census tract code identifying the census tract within the county.
<code>blkgrpce</code>	Single-digit block-group code identifying the block group within the census tract.
<code>geom</code>	Spatial geometry column containing the boundary polygons for each census unit, in the form of an <code>sf</code> geometry object.

Core Area CBSA/CSA Prebuild Shapefiles

For the implementation, refer to **SUBSECTION A4: Build Precompiled TIGER/Line GeoPackages** in “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. As described in the methods for LOOP PART D: Point-in-Polygon Spatial Assignment of Census Information, pulling in shapefiles using the `tigris` package was computationally prohibitive. Instead, relevant

state-level shapefiles for the Metropolitan and Micropolitan Statistical Areas (CBSA) and Combined Statistical Areas (CSA) were manually downloaded from the Census Bureau TIGER/Line Shapefiles database (see [Directory Specific Notes](#) in the project GitHub repository for direct links to sources) ⁴. These were then compiled into standardized shapefiles with layers for each decennial period (2000, 2010, and 2020) and written to the `./Data/Results/Census Bureau TIGER Line Shapefiles` directory.

Table 6: Codebook for Core Area Shapefiles (CBSA/CSA; 2007/2010/2020 Layers)

Field	Description
<code>vintage_year</code>	The decennial census year associated with the boundary definition (2007, 2010, or 2020).
<code>area_type</code>	Classification of the area as either a Core-Based Statistical Area (CBSA) or a Combined Statistical Area (CSA).
<code>area_code</code>	Official Census Bureau numeric code uniquely identifying the CBSA or CSA.
<code>area_level</code>	Hierarchical classification of the CBSA, indicating whether the area is Metropolitan or Micropolitan; CSA entries are NA.
<code>area_name</code>	Full official name of the statistical area as designated by the Census Bureau (e.g., Atlanta-Sandy Springs-Alpharetta, GA).
<code>area_states</code>	Concatenated string of state abbreviations for all states encompassed by the statistical area boundary.
<code>geom</code>	Spatial geometry column containing the boundary polygons for each census unit, in the form of an <code>sf</code> geometry object.

Core Area ZCTA Prebuild Shapefiles

For the implementation, refer to **SUBSECTION A4: Build Precompiled TIGER/Line GeoPackages** in [“Code/Clean Raw Data_Step 2_2026 Format.R”](#). As described in the methods for LOOP PART D: Point-in-Polygon Spatial Assignment of Census Information, pulling in shapefiles using the `tigris` package was computationally prohibitive. Instead, relevant state-level shapefiles for the ZIP Code Tabulation Areas (ZCTA) were manually downloaded from the Census Bureau TIGER/Line Shapefiles database (see [Directory Specific Notes](#) in the project GitHub repository for direct links to sources) ⁴. These were then compiled into standardized shapefiles with layers for each decennial period (2000, 2010, and 2020) and written to the `./Data/Results/Census Bureau TIGER Line Shapefiles` directory.

Table 7: Codebook for ZCTA Shapefiles (2000/2010/2020 Layers)

Field	Description
vintage_year	The decennial census year associated with the boundary definition (2000, 2010, or 2020).
area_type	Classification of the area as ZCTA.
area_code	Official Census Bureau numeric code uniquely identifying the ZCTA.
area_states	Concatenated string of state abbreviations derived from a spatial join with the state-level TIGER/Line Shapefile for the respective decennial year.
geom	Spatial geometry column containing the boundary polygons for each census unit, in the form of an sf geometry object.

Step 2 Final Result

For the implementation, refer to “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. The standardized records generated in **Step 1** were run through the sequential cleaning and validation steps outlined in **Step 2**. Results were compiled across a batch’s array outputs and aggregated between batches into a single complete file in **PART C: Recompile Results from the HPC**, then written to “./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/church_2026_form_validated_08.03.2026.db”.

Table 8: Codebook for new output fields produced by the evaluation.

Field	Description
address	Best-available address: verified_address (trimmed; excluding “No address match found”), else matched_address, else combined_address.
address_verified	USPS-verified address result when available; NA if no match is found or verification did not succeed.
address_matched	Result from agnostic address matching when performed (e.g., exact or fuzzy match). “Only one address” indicates matching was attempted but only one address was reported.
reported_address	Original address as reported, prior to cleaning or verification.
longitude/latitude	Best-available coordinates: verified coordinates when available, otherwise averaged raw coordinates (stored as numeric).
geolocation_verified	Indicator that geocoordinates were verified (definition depends on the verification step used).
geoid_[2000 2010 2020]	Full census GEOID assigned via point-in-polygon spatial join with block-group TIGER/Line boundaries for the corresponding decennial year.
geoid_match	Summary of GEOID assignment across years: “Matched” (all years), “Some matches not found” (partial), “Matches not found” (none), or “Not enough info” (insufficient coordinate data).
cbsa_code_[2007 2010 2020]	CBSA numeric code assigned via point-in-polygon spatial join for the corresponding vintage year.
cbsa_level_[2007 2010 2020]	CBSA level (Metropolitan vs. Micropolitan) for the corresponding vintage year.
csa_code_[2007 2010 2020]	CSA numeric code assigned via point-in-polygon spatial join for the corresponding vintage year.
zcta_[2000 2010 2020]	Five-digit ZCTA assigned via point-in-polygon spatial join with ZCTA TIGER/Line boundaries for the corresponding decennial year.

Step 2 Final Result - Import Quality Checks

For the implementation, refer to “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. The standardized records generated in **Step 1** were run through the sequential cleaning and validation steps outlined in **Step 2**. Results were compiled across a batch’s array outputs and aggregated between batches into a single complete file in **PART C: Recompile Results from the HPC**.

Selected outcomes and characteristics of these within-batch imports were generated to enable rapid confirmation of expected coverage and completeness. These were written with the main results to “./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/church_2026_form_validated_08.03.2026.db. Each batch’s results are annotated with the first two digits of the batch number (<XX>).

Table 9: Codebook for import_qc_<XX>\$abi_check

Field	Description
array	Batch array number from which the results were generated.
file	Source file name for the batch results.
from/to	ABI search-space index range defined under “SUBSECTION B1: Index Queue” in Clean Raw Data_Step 2 HPC v2_2026 Format.R.
n_expected_unique	Number of unique ABIs expected from the Step 1 data for the given index range.
n_actual_unique	Number of unique ABIs present in the batch results for the given index range.
n_missing	Number of ABIs absent from the batch results, computed as $ \text{setdiff}(\text{expected}, \text{actual}) $.
qc_pass	Boolean: TRUE if $n_{\text{missing}} = 0$, otherwise FALSE.

Table 10: Codebook for `import_qc_<XX>${address_matched|address_verified|geoid_match|geolocation_verified}`

Field	Description
array	Batch array number from which the results were generated.
value	Unique outcome observed for the QC column (e.g., "Exact match").
n	Number of records with the given outcome.
n_addr	Total number of addresses in the array.
pct	Percentage of addresses with the given outcome.

Table 11: Codebook for `import_qc_<XX>${na_census_boundaries}`

Field	Description
array	Batch array number from which the results were generated.
column	Variable checked for NA values.
n_addr	Total number of addresses in the array.
n_na	Number of NA entries observed for the column.
pct_na	Percentage of addresses with an NA for that census boundary column.

Step 2 Algorithm Quality Check Datasets - Import Quality Checks

For the implementation, refer to “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. The three quality control datasets — `address_qc`, `census_qc`, `geo_qc` — collected at the time of import were compiled from multiple HPC batches. A series of quality checks were run to confirm ABI coverage against the **Step 1** data. Each batch’s results are annotated with the first two digits of the batch number (<XX>). These were written with the main results to the `./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/` directory.

Table 12: Codebook for `import_qc_<XX>`

Field	Description
array	Batch array number from which the results were generated.
file	Source file name for the batch results.
key	Quality-check list (e.g., qc1, qc2) within the file from which the results were generated.
from/to	ABI search-space index range defined under “SUBSECTION B1: Index Queue” in Clean Raw Data_Step 2 HPC v2_2026 Format.R.
n_expected_unique	Number of unique ABIs expected from the Step 1 data for the given index range.
n_actual_unique	Number of unique ABIs present in the batch results for the given index range.
n_missing	Number of ABIs absent from the batch results, computed as $ \text{setdiff}(\text{expected}, \text{actual}) $.
qc_pass	Boolean: TRUE if $n_{\text{missing}} = 0$, otherwise FALSE.

Step 2 Algorithm Quality Check Datasets - Addresses

For the implementation, refer to “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. The three quality control (QC) datasets — address_qc, census_qc, geo_qc — collected at the time of import were compiled from multiple HPC batches. The address QC dataset included three checks covering the address validation and matching process:

- **QC1:** A general indicator of whether API verification was allowed, which ABIs were verified, whether any addresses were matched, and two checks for duplications or other introductions and losses of results.
- **QC2:** Summarizes all quality check outcomes from address validation using the USPS API over unique ABIs and addresses.
- **QC3:** Summarizes all quality check outcomes from address matching over unique ABIs and addresses.

These were written with the main results to the “./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/address_qc_08.05.2026.db

directory.

Table 13: Codebook for `qc1`

Field	Description
<code>i</code>	Index of the <code>for</code> loop iteration (results are stored as lists).
<code>abi</code>	Business ID to which the QC results pertain.
<code>allow_usps_api</code>	Boolean: TRUE if USPS API usage was permitted during that array run via <code>verify_addresses</code> ; otherwise FALSE.
<code>api_used</code>	Boolean: TRUE if the ABI entries are configured for USPS API usage via <code>do_api</code> ; otherwise FALSE.
<code>verification_attempted</code>	Boolean: TRUE if <code>verified_address</code> is present. This reiterates <code>do_api</code> and should match it.
<code>match_attempted</code>	Boolean: TRUE if agnostic matching was attempted for any addresses for that ABI, indicated by the presence of <code>matched_address</code> .
<code>duplicates_induced</code>	Boolean: TRUE if any duplicates over <code>combined_address</code> , <code>anchor_year_min</code> , and <code>anchor_year_max</code> are detected; otherwise FALSE.
<code>any_addresses_line1_na</code>	Boolean: TRUE if any <code>address_line_1</code> for that ABI is NA; otherwise FALSE.
<code>new_not_in_old_addr_yr</code>	From <code>compare_tabs()</code> : indicates whether the cleaned data contains any unique <code>combined_address</code> and <code>archive_version_year</code> outcomes not present in the Step 1 standardized data.
<code>old_not_in_new_addr_yr</code>	Inverse of <code>new_not_in_old_addr_yr</code> : indicates whether the Step 1 standardized data contains any outcomes not present in the cleaned data.
<code>new_vs_old_differ</code>	Boolean: TRUE if the processed data differs in any way from the Step 1 standardized form; otherwise FALSE.

Table 14: Codebook for qc2

Field	Description
i	Index of the for loop iteration (results are stored as lists).
abi	Business ID to which the QC results pertain.
address	Best-available address: <code>verified_address</code> (trimmed; excluding "No address match found"), else <code>matched_address</code> , else <code>combined_address</code> .
archive_versions_present	String listing all years in which the address was observed.
reported_address	Original reported address, prior to any cleaning or verification.
address_verified	If the address was verified against the USPS API, this field is TRUE. If unverified but a match is found, the value indicates the string-matching mode ("Override" or "Exact"). If no <code>address_line_1</code> was present for matching, this is indicated with "No <code>address_line_1</code> ".
ver_geolocation_test	Geolocation test for matching attempts to verified addresses: "Override" if "Exact" matched; TRUE/FALSE if "Fuzzy" matched; NA if no match was attempted.
usps_status	Numeric status code summarizing the USPS API query interaction (e.g., 200, 400).
usps_status_detail	Status description corresponding to <code>usps_status</code> (e.g., "200 Successful operation").
attempt_succeeded	Indicates whether verification succeeded on the first try, succeeded via fallback, or failed entirely.
verified_address	Verified address returned by the USPS API query.

Table 15: Codebook for qc3

Field	Description
i	Index of the for loop iteration (results are stored as lists).
abi	Business ID to which the QC results pertain.
address	Best-available address: <code>verified_address</code> (trimmed; excluding “No address match found”), else <code>matched_address</code> , else <code>combined_address</code> .
archive_versions_present	String listing all years in which the address was observed.
reported_address	Original reported address, prior to any cleaning or verification.
address_matched	If agnostic address matching was performed, this field records the outcome when matching succeeded via “Exact matching” or “Fuzzy matching”. “Only one address” indicates matching was attempted but only one address was reported.
match_geolocation_test	Geolocation test for agnostic matching attempts: “Override” if “Exact” matched; PASS/FAIL if “Fuzzy” matched. If PASS includes “no Lon”, “no Lat”, or “no Lon/Lat”, one record was missing that coordinate pair. NA if no match was attempted.
matched_address	Matched address result.

Step 2 Algorithm Quality Check Datasets - Census

For the implementation, refer to “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. The three quality control (QC) datasets — `address_qc`, `census_qc`, `geo_qc` — collected at the time of import were compiled from multiple HPC batches. **QC1** and **QC2** apply the same tests over different columns, verifying whether the reported census boundaries correspond with any of the spatial join results, and whether the matched vintages align with the dates the address was filed.

These were written with the main results to the “./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/census_qc_08.06.2026.db” directory.

Table 16: Codebook for qc1

Field	Description
i	Index of the for loop iteration (results are stored as lists).
abi	Business ID to which the QC results pertain.
address	Best-available address: <code>verified_address</code> (trimmed; excluding "No address match found"), else <code>matched_address</code> , else <code>combined_address</code> .
archive_versions_present	String listing all years in which the address was observed.
census_[block tract]/ [county fips]_code	Original reported census boundaries recorded for the address (block/tract and county/FIPS codes, as available).
n_address	Number of times the abi-address combination was recorded.
geoid_[2000 2010 2020]	Full concatenated census GEOID assigned via point-in-polygon spatial join with block-group TIGER/Line boundaries for the corresponding decennial year.
[fips county tract block] _code_any_match	Whether any original reported value matches any decennial value annotated via spatial joining: "Matched" if any do; "Uncheckable" if no original value was available; "None" if no matches.
[fips county tract block] _vintages	Which decennial vintages matched: "None" if none matched, or "Not reported" if no original value was reported.
[fips county tract block] _vintages_aligned	From <code>check_alignment()</code> : TRUE if matched vintages align with the years the address was reported; FALSE otherwise; NA if no comparison was available.
block_type	Level of the reported block boundary: "None reported" if missing; "Blocks" if four digits; "Block groups" if one digit; or "Not the expected dimensions" otherwise.

Table 17: Codebook for qc2

Field	Description
<code>i</code>	Index of the <code>for</code> loop iteration (results are stored as lists).
<code>abi</code>	Business ID to which the QC results pertain.
<code>address</code>	Best-available address: <code>verified_address</code> (trimmed, excluding "No address match found"), else <code>matched_address</code> , else <code>combined_address</code> .
<code>archive_versions_present</code>	String listing all years in which the address was observed.
<code>n_address</code>	Number of times the <code>abi</code> -address combination was recorded.
<code>cbsa_[level code]/csa_code</code>	Original reported CBSA/CSA boundaries for the address (CBSA level and/or CBSA code, and CSA code, as available).
<code>cbsa_code_[2007 2010 2020]</code>	Official Census Bureau CBSA numeric code assigned via point-in-polygon spatial join for the corresponding vintage year.
<code>cbsa_level_[2007 2010 2020]</code>	CBSA classification (Metropolitan vs. Micropolitan) for the corresponding vintage year.
<code>csa_code_[2007 2010 2020]</code>	Official Census Bureau CSA numeric code assigned via point-in-polygon spatial join for the corresponding vintage year.
<code>cbsa_[level code] _code_any_match /csa_code_code_any_match</code>	Whether any original reported value matches any vintage value annotated via spatial joining: "Matched" if any do; "Uncheckable" if no original value was available; "None" if no matches.
<code>cbsa_[level code]_vintages/ csa_code_vintages</code>	Which vintages matched: "None" if none matched, or "Not reported" if no original value was reported.
<code>cbsa_[level code]_vintages _aligned/ csa_code_vintages_aligned</code>	From <code>check_alignment()</code> : TRUE if matched vintages align with the years the address was reported; FALSE otherwise; NA if no comparison was available.

Step 2 Algorithm Quality Check Datasets - Geocoordinates

For the implementation, refer to “[Code/Clean Raw Data_Step 2_2026 Format.R](#)”. The three quality control (QC) datasets — `address_qc`, `census_qc`, `geo_qc` — collected at the time of import were compiled from multiple HPC batches. Three quality checks were conducted over the geocoordinate verification process:

- **QC1:** Summarizes all quality check outcomes from address-based geocoding validation using the US Census Bureau over unique ABIs and addresses.
- **QC2:** Assesses the difference between the reported and validated geocoordinates.
- **QC3:** Examines the spread of geocoordinates among all records sharing the same ABI and address, and indicates whether this summary includes records where the address was verified or agnostically matched.

These were written with the main results to the `"/Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/geo_qc_08.06.2026.db` directory.

Table 18: Codebook for `qc1`

Field	Description
<code>i</code>	Index of the <code>for</code> loop iteration (results are stored as lists).
<code>abi</code>	Business ID to which the QC results pertain.
<code>address</code>	Best-available address: <code>verified_address</code> (trimmed; excluding “No address match found”), else <code>matched_address</code> , else <code>combined_address</code> .
<code>archive_versions_present</code>	String listing all years in which the address was observed.
<code>n_address</code>	Number of times the <code>abi</code> -address combination was recorded.
<code>enough_geo</code>	Boolean: TRUE if both longitude and latitude were reported; otherwise FALSE.
<code>geolocation_verified</code>	Boolean: TRUE if an address-based geocoding match was retrieved from the US Census Bureau Geocoder API; otherwise FALSE.

Field	Description
<code>n_attempts</code>	Number of API query attempts made; each attempt uses a different benchmark and vintage in the query URL.
<code>query_statuses</code>	Summary of returned query statuses. 200 indicates a successful interaction even if no match was found. Attempts are separated by a vertical bar.
<code>all_200</code>	Boolean: TRUE if all benchmark/vintage pairs returned status 200 (interaction succeeded but no match found); otherwise FALSE.
<code>geo_matched_address</code>	Matched address returned by the geocoding database.
<code>matched_address_same</code>	Uses <code>find_similar_addresses(threshold = 0)</code> to check whether the addresses are exactly the same.
<code>matched_address_similar</code>	Uses <code>find_similar_addresses(threshold = 0.15)</code> to check whether the addresses are similar.
<code>benchmark/ vintage_input</code>	Specific reference database (benchmark and vintage) used to search for an address match during geocoding.
<code>address_verified</code>	If the address was verified against the USPS API, this field is TRUE. If unverified but a match is found, the value indicates the string-matching mode ("Override" or "Exact"). If no <code>address_line_1</code> was present for matching, this is indicated with "No <code>address_line_1</code> ".
<code>ver_geolocation_test</code>	Geolocation test for matching attempts to verified addresses: "Override" if "Exact" matched; TRUE/FALSE if "Fuzzy" matched; NA if no match was attempted.
<code>address_matched</code>	If agnostic address matching was performed, this field records the outcome when matching succeeded via "Exact matching" or "Fuzzy matching". "Only one address" indicates matching was attempted but only one address was reported.
<code>match_geolocation_test</code>	Geolocation test for agnostic matching attempts: "Override" if "Exact" matched; PASS/FAIL if "Fuzzy" matched. If PASS includes "no Lon", "no Lat", or "no Lon/Lat", one record was missing that coordinate pair. NA if no match was attempted.

Table 19: Codebook for qc2

Field	Description
i	Index of the for loop iteration (results are stored as lists).
abi	Business ID to which the QC results pertain.
address	Best-available address: <code>verified_address</code> (trimmed; excluding “No address match found”), else <code>matched_address</code> , else <code>combined_address</code> .
archive_versions_present	String listing all years in which the address was observed.
[lat lon]_abs_diff	The absolute difference between the reported geocoordinate and the validated one.
[lat lon]_gt_002	Boolean: TRUE if the absolute difference exceeds 0.002 degrees; otherwise FALSE. NA if there were no geocoordinates to compare.

Table 20: Codebook for qc3

Field	Description
i	Index of the for loop iteration (results are stored as lists).
abi	Business ID to which the QC results pertain.
address	Best-available address: <code>verified_address</code> (trimmed; excluding “No address match found”), else <code>matched_address</code> , else <code>combined_address</code> .
n_address	Number of times the abi-address combination was recorded.
[lat lon]_ [min q1 median mean q3 max]	Summary statistics describing the spread of geocoordinates reported for a unique address.
[lat lon]_spread_gt_02	Boolean: TRUE if the spread of geocoordinates exceeds 0.02 degrees; otherwise FALSE. NA if there were no geocoordinates to compare.

Field	Description
any_address_verified	Boolean: TRUE if any address in these results came from a verified address; NA if no validation was used.
any_address_matched	Boolean: TRUE if any address in these results came from an agnostically matched address; NA if no validation was used.

References

1. Ushey, K., Wickham, H. & Posit. [Introduction to renv](#).
2. Walker, K. & Rudis, B. [Tigris: Load census TIGER/line shapefiles](#). (2025).
3. SimpleMaps. United states cities database. <https://simplemaps.com/data/us-cities>.
4. U.S. Census Bureau. TIGER/line shapefiles. Census.gov. <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html> (2025).